5.1 Machin's Formula

WORKSHEET

The question

STEP III 1987, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What is the argument of $5 - \mathrm{i}$? Of $1 + \mathrm{i}$? What happens to arguments when complex numbers are multiplied?

STAGE 1 Bounds on the modulus

Covers: Part (i)

Ask yourself For the upper bound, square both sides. Both are non-negative.

Pause and try For the lower bound, think of $|x|\cos\alpha + |y|\sin\alpha$ as the dot product of $(|x|, |y|)$ with a unit vector. How big can a dot product be?

STAGE 2 The key multiplication

Covers: Part (ii), the first formula

Ask yourself When complex numbers are multiplied, the arguments add. What are the arguments of $5 - \mathrm{i}$ and $1 + \mathrm{i}$? Then work out $(5 - \mathrm{i})^4(1 + \mathrm{i})$ step by step.

Pause and try What is the argument of your answer? Use part (i), or the sizes of the angles, to show there is no extra multiple of 2π.

STAGE 3 The second formula

Covers: Part (ii), the second formula

Ask yourself By analogy, which product of complex numbers has argument $3\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{1}{20}+\tan^{-1}\frac{1}{1985}$? Multiply it out.

What's really going on

Computing π by hand

Why is Machin's formula so good for computation? The series $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ converges for $|x| \le 1$, but painfully slowly at $x = 1$: you need about a thousand terms of $1 - \frac{1}{3} + \frac{1}{5} - \cdots$ for three decimal places of $\frac{\pi}{4}$. With $x = \frac{1}{5}$, each term is about 25 times smaller than the last, giving roughly 1.4 decimal digits per term; with $x = \frac{1}{239}$, about 4.8 digits per term. Machin's 100 digits needed only about 70 terms of the first series and 20 of the second. In 1873 William Shanks used a Machin formula to compute 707 digits, spending years on the arithmetic; in 1944 it was discovered that he had made an error at the 528th place, and everything after it was wrong.

Gaussian integers again

The numbers $a + b\mathrm{i}$ with a, b whole numbers are the *Gaussian integers* of Section 1.6. The reason $5 - \mathrm{i}$ works so well is a factorisation: $239^2 + 1 = 57122 = 2 \times 13^4$, which in the Gaussian integers reads $(239 + \mathrm{i})(239 - \mathrm{i}) = (1 + \mathrm{i})(1 - \mathrm{i})\,(3 + 2\mathrm{i})^4(3 - 2\mathrm{i})^4$, and $5 - \mathrm{i} = (1 - \mathrm{i})(3 + 2\mathrm{i})$ is built from the same Gaussian primes. Finding Machin-type formulas is really a problem about factorising numbers of the form $n^2 + 1$. In 1899 Carl Størmer used exactly this idea to prove that there are only four formulas of the form $\frac{\pi}{4} = a\tan^{-1}\frac{1}{m} + b\tan^{-1}\frac{1}{n}$ with whole numbers a, b, m, n: Machin's, and three found by Euler, Hermann and Hutton.

Rotations

Multiplying by a complex number of modulus 1 is a rotation of the plane. That is why complex numbers are the natural language for anything involving rotation and angles, from the geometry later in this chapter to alternating currents in electrical engineering

and the wave functions of quantum mechanics. In Section 4.2 the addition formula for $\tan^{-1}$ came from a clever substitution; here it comes from the multiplication of complex numbers, and the two are the same fact seen from different sides.

Try this next

1. **A cousin.** Try STEP II 2000, Question 4 (step.maths.org/questions/00-s2-q4). It proves another arctangent formula for $\frac{\pi}{4}$ in the same way as this section: multiply complex numbers and read off the argument.

2. **Euler's formula.** Show that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4}$ by considering $(2+\mathrm{i})(3+\mathrm{i})$.

3. **Hutton's formula.** Show that $\frac{\pi}{4} = 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7}$.

4. **How many terms?** Estimate how many terms of each series in Machin's formula you would need to compute π to 20 decimal places. Then do it for the first few digits with a calculator.

Further reading

- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). Includes the history of Machin-type formulas.
- **Jörg Arndt & Christoph Haenel, *Pi Unleashed*** (Springer). A lively account of methods for computing π, old and new.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). The best book for seeing complex numbers geometrically. Chapter 1 is about multiplication as rotation and scaling.

The full guided solution is in Section 5.1:
https://beyondstep.keithmcnulty.org/ch05/machin.html